UK CanSat Competition

Team name: CERC

School: Cambourne Village College

Preliminary Design Review

Date: 21/11/25

1 INTRODUCTION

1.1 Team Organisation and Roles

Team Lead - Organise the members, tasks and building of the Cansat, ensure that deadlines are being met and complete most documentation

Software design lead - Create and manage software as per required for the CanSat, working closely with the electrical design lead.

Electrical design lead - Design the necessary circuitry and use relevant microcontrollers and sensors to create necessary systems for the Cansat.

Mechanical design lead - Use CAD, sketches and other relevant design tools to engineer the Cansat's structure. Works closely with the electrical design lead.

Landing and ground systems lead - Ensure the reliability of the ground telecommunication systems and designs the landing system

Testing lead - Design experiments to test components and the Cansat as a whole, carry these out and record relevant data to allow other members to make improvements

1.2 Mission Overview

1.2.1 Mission Objectives

Our main objectives are:

- To measure temperature and air pressure to calculate altitude and demonstrate correlations between these values
- To analyse the aerodynamic behaviour of our CanSat
- To collect scientific data that is sent to the ground station and later analysed

Primary Mission:

The primary mission of our Cansat is to measure air pressure and temperature using sensors, and transmit this as telemetry data every second to our ground station. We will calculate altitude directly from air pressure and send this data as well. The received data will be formatted into a CSV file which we can then analyse and plot. More detail on this is given in the software design section.

Secondary Mission:

For our secondary mission, we will use a sensor that acts as an accelerometer and gyroscope to work out relevant aerodynamic behaviour of our CanSat. We will work out the level of turbulence using the accelerometer. We will also measure the oscillation and rotation of our CanSat with our gyroscope, and use that to evaluate the stability of our parachute.

We hypothesise that turbulence will decrease as altitude increases due to fewer convection currents, and that there will be some oscillation with our parachute but it will be mostly stable.

1.2.2 What will you measure, why and how?

For the primary mission:

- To control the activities of our sensors we will use a Pi Pico as our microcontroller
- To measure air pressure and temperature we will use a BMP-280
- To send and receive radio signals we will use two Adafruit LoRa radio modules and a Yagi antenna
- To program the sensors we will be using CircuitPython
- We will calculate the terminal velocity of our CanSat by looking at the rate of descent of our altitude measurements
- We can also calculate the drag coefficient of our parachute. We simply use the fact that when the CanSat reaches its terminal velocity, the force of drag is equal to the weight force of the CanSat. Then we can do similar calculations to those we used to calculate the size of our parachute to work out the drag coefficient.

For the secondary mission:

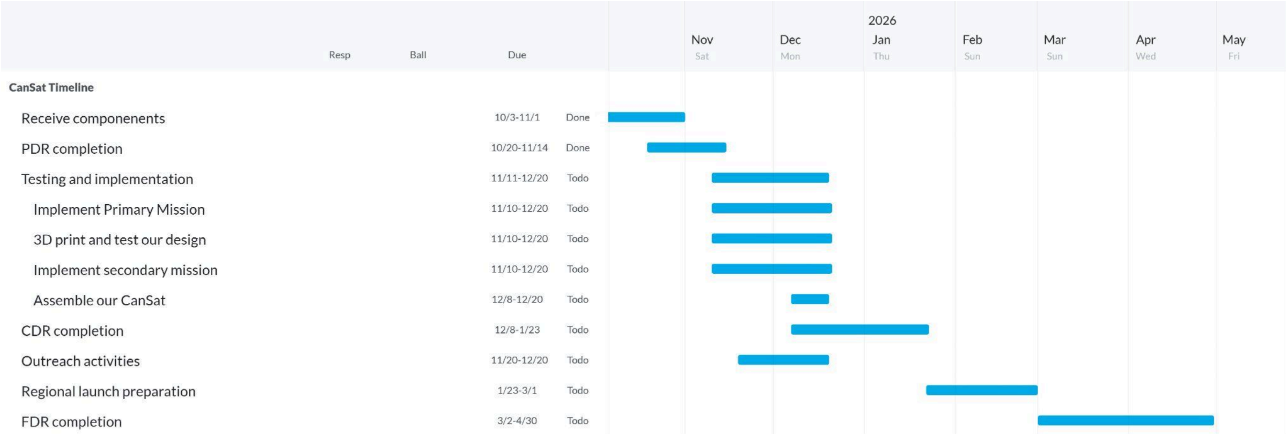
- To calculate the turbulence of our CanSat we will use the MPU6050 accelerometer. As it returns acceleration in all three axes, we can ignore the vertical axis of acceleration, and look at the two remaining axes. By taking the variance in this acceleration data in certain sections of our flight, we can use this as a measure of turbulence.
- To calculate the oscillation of our CanSat, we can look towards our gyroscope. As it returns values in degrees per second, we will analyse this data. To give a measure of our CanSat's oscillation, we will consider a few factors. A constant spin is unavoidable and should not be penalized heavily, so our calculation will involve the variance in our data. However, an extreme rate of rotation is also unwanted, so we multiply by the mean of our data, punishing a high overall rate of spin, and also multiply by the peak rate of change to punish any small sections of extremes. We also want to punish sudden jerks, so we multiply by the mean angular acceleration. We will likely scale each individual value between 0 and 1 to keep the end value in this range as well. This creates a holistic measure of oscillation that can easily be used to evaluate the effectiveness of our parachute.
- To analyse and plot data we will use the python libraries numpy, pandas and matplotlib.

To measure turbulence we ideally want a high sampling rate, but the radio module we are using may not allow us to transmit data at such a rate. Therefore, we may also consider storing more data internally on the CanSat, to allow for even better measurements.

More details around choices of hardware and software are explained in their respective sections.

2 Project Planning

2.1 Time schedule



We plan to split the work for our CanSat build evenly between our team, so that we can work in parallel, completing many tasks at a faster rate. This will also allow our team members excell in their respective strengths. This is preferred to all of us working on one project at a time, as those without experience will be sitting idle whilst those with experience will take extra workloads.

We have broken up the main tasks into various subtasks, and divided them between people. This approach started with the PDR, where based on their roles, we tasked specific people with specific sections, based on their areas of expertise.

For the assembly of our CanSat, we plan to, as aforementioned, split the work into multiple tasks that can be completed in parallel. These will ideally be completed in pairs.

For the primary mission, we will first get the BMP-280 to give accurate readings of temperature and pressure, and calculate altitude from the pressure. We will then implement the telemetry. For our CanSat's design we will 3D print our design and test its strength as described in the testing section. As for the secondary mission, there are again two parts. First we need to get the sensor to work and test its accuracy, and then we need to design a program that will use these values to calculate the CanSat's position, and test its accuracy.
Towards the end of the time allocated to implementation in our Gantt chart, after all of these separate tasks have been completed,we will put all of these components together and assemble our CanSat. This will then be tested.

After this we can begin work on our CDR. For this we will follow a similar divide and conquer approach to that which we used in our PDR, assigning individuals certain sections. This will resemble a more detailed version of our PDR, elaborating on ideas, explaining finalised choices and showing the work that we have done in between the design reports. After the CDR our focus will shift entirely towards refining and testing our CanSat to ensure that it is ready for the regional launch

If we are accepted into the nationals, our focus will then of course shift towards the FDR and national launches. We will focus more on the FDR than the launches as this is where the most improvements will need to be made. We will follow the same divide and conquer yet again, but this time learning from our experiences in the regional launches and our feedback from the PDR and CDR.

2.2 Team and External Support

	Programming	Electronics	Mechanical design	
Student 1 (Team Lead)	Excellent programming skills specifically with micropython	Substantial experience with electronics	Some skills and has good ideas	Lots of experience with public speaking and excellent leadership skills
Student 2	Excellent programming skills, extensive experience with micropython	Substantial practical experience with electronics	Few skills but has good ideas	Some experience with public speaking and good leadership skills
Student 3	Excellent programming skills with microcontrollers	Lots of applied practical experience with electronics	Very good ideas and experience designing	Some experience with public speaking and confident leadership
Student 4	Good programming skills especially with micropython, but may require more support	Some experience with electronics	Excellent ideas and plenty of experience designing for several purposes	Experience with public speaking and leadership
Student 5	Little programming skills, not his area of expertise	Little experience with electronics, but could be taught	Would come up with plenty of ideas and some experience	Experience presenting and some leadership experience
Student 6	Some basic programming skills, requires assistance from peers	Little experience with electronics, but could potentially be taught	Good ideas and some experience	Experience presenting and confident in leadership experience

Best Skill Strong Skill Secure Skill Weak Skill

Overall, we believe that we possess a well-balanced skillset as a team, with strong skills across key areas which are important for the project. These skills map well to our team roles, meaning everyone is in a position well-suited for them. Team members have sufficient skills in areas outside of their expertise to support others in their responsibilities, and we have enough time to teach weaker members so they can also be of assistance.

However, despite our well-balanced skill set there are still some areas we will likely struggle with. Although our team leader is very experienced in outreach, other team members may lack the confidence to run activities such as workshops or school assemblies. To solve this, we will slowly build up to the most challenging activities by doing activities that are less interactive or with smaller groups of people first, allowing students to gain the necessary confidence. We will also work closely with our local robotics club and potentially have other members of the club or staff assist with the activities we are running. Another key area we may require support with is soldering. Although our team has very good experience with robotics, we do not have much experience with soldering. As good soldering is vital to the reliability of our CanSat, as our team's lack of experience could cause major issues. We will need to work on this skill and/or seek external support by getting someone to teach us, likely from our robotics club.

If there are any other areas in which we require support, we will seek it from our mentor, the competition guidelines and friends who have participated in this competition in the past.

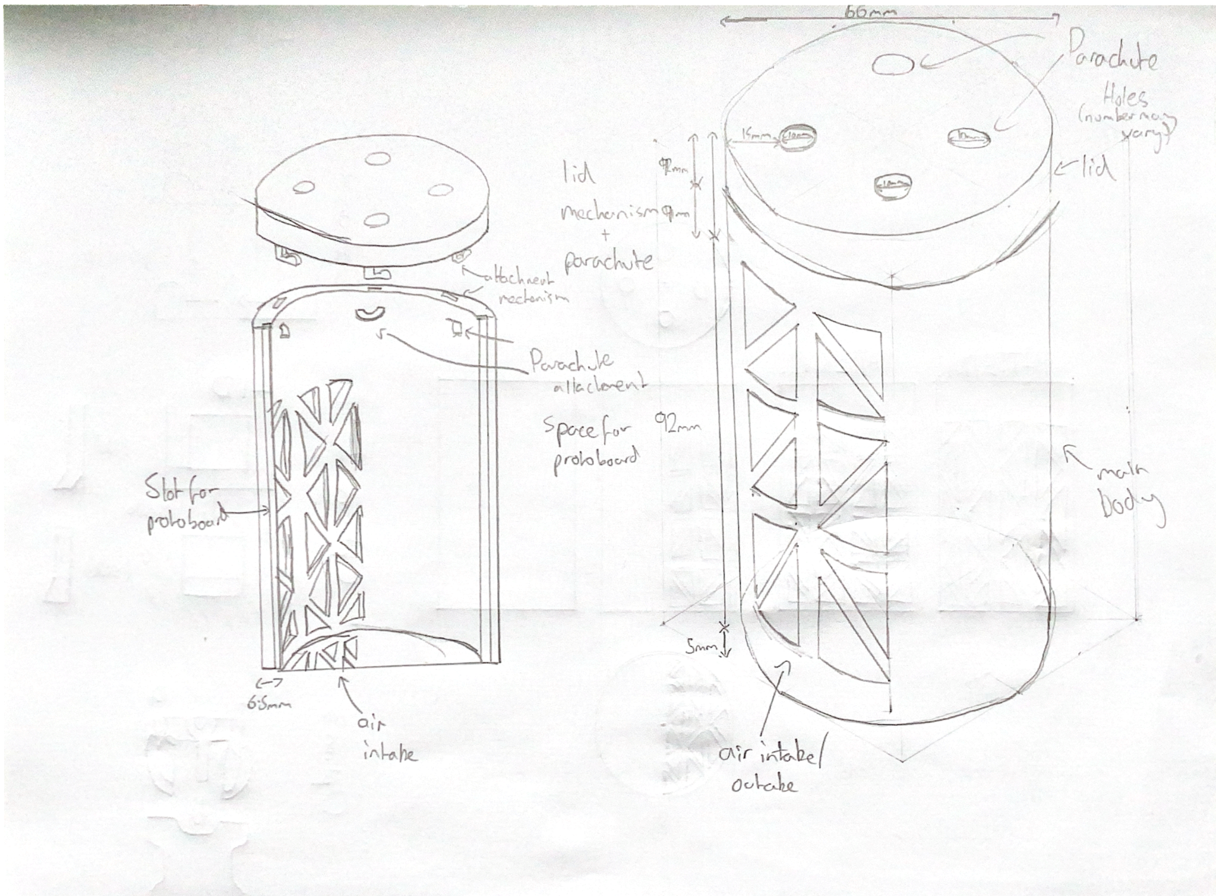
2.3 Risk Analysis

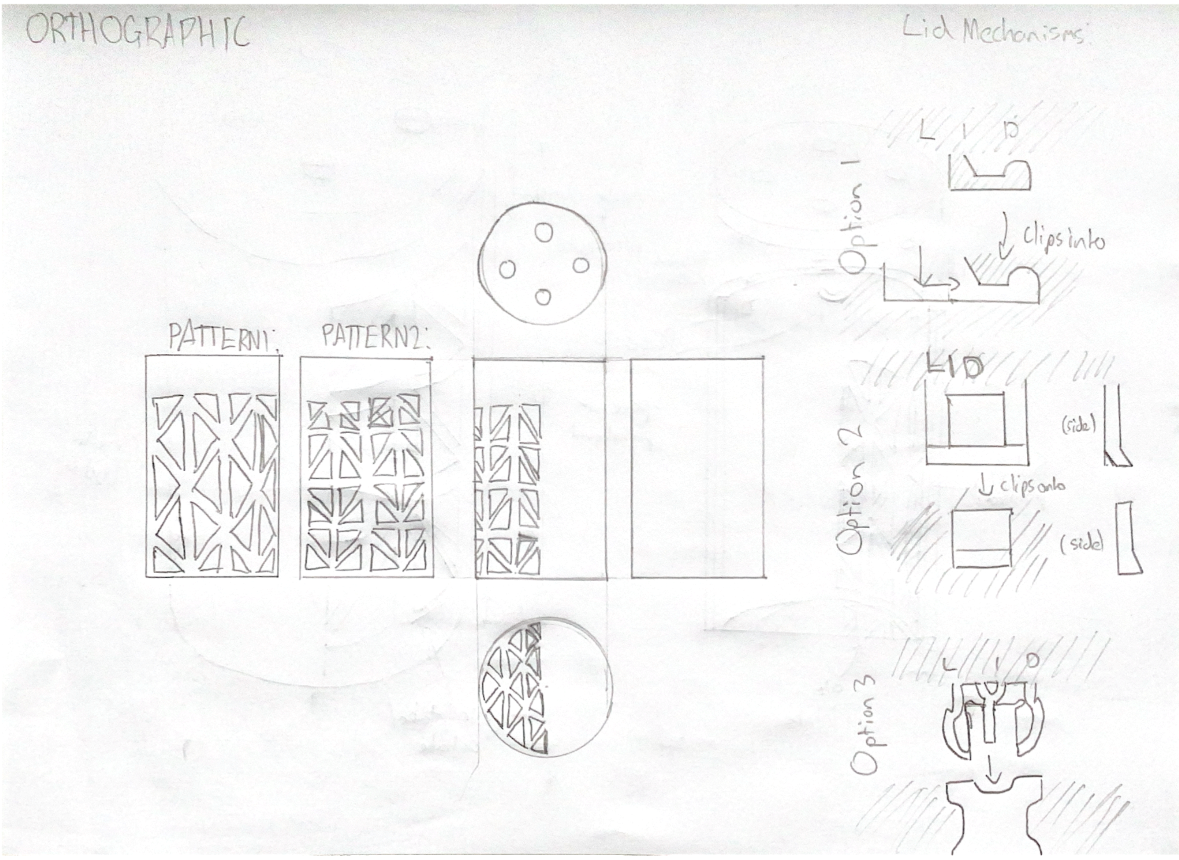
We have identified the overarching risks that would pose major issues for the project. However, as we are very early in project development , many unexpected issues which we cannot plan for at this stage will likely arise, so we will have to face these issues as they arrive

Risk	Potential consequences	Potential mitigations
Team management difficulties/skill gaps	Could lead to poor documentation and designs if team members are unclear on their task, are assigned to a task they are not comfortable with, or do not have a task to do. This will contribute to the probability of technical failures on the launch day and in general. The team would not be being utilised to its full potential.	We will use rigid planning structures, giving each member a specific role and our team leader will assign each role separate tasks to work on according to their strengths and weaknesses, as outlined in previous sections. Where there are skill gaps, we will ensure that external support is provided. Our team leader reviews all important changes to documentation, ensuring quality and providing support if standards are not met.
Time constraints	Possibility of unfinished or poor quality documentation as deadlines are not being met, critical oversights in design leading to preventable failures of the CanSat, or unfinished assembly of the CanSat before launch day, which has happened to teams in previous years.	Much like the previous risk, we will attempt to mitigate this with strict planning schedules. We will aim to have important documentation started as soon as possible before the deadline, and have a working draft soon after. CanSat assembly, such as 3D printing, will be done as soon as the applicable designs and programs are ready, to ensure there is ample time for testing.
Equipment failures	Unreliable equipment at the ground station ,such as a poorly assembled antenna, or issues like poor soldering inside the CanSat itself, will likely cause us to lose some or all of our data, negating much of the work put in and giving a disappointing outcome to the launch.	As long as time scheduling is followed as above, we should have a working version of the ground system and CanSat months before the launch. We will use this time to ensure everything is built as robustly as possible, and carry out extensive testing on all important systems. Depending on the conclusions made from this , we will make necessary adaptations to the design, prioritising reliability and functionality over more advanced features.

3 CANSAT DESIGN

3.1 Mechanical design





Design

The CanSat's design is built around durability, so as to prevent component failure and survive the impact upon landing. Thick walls will have grooves cut into them for the protoboard to be inserted vertically, and the battery fitted behind it, using a bracket to hold it in place. On the other side of the protoboard, ventilation holes in the walls and underside of the CanSat will allow for increased airflow and more accurate readings from sensors. Multiple patterns for the holes have been put forward, and will be extensively tested for their strength and ventilation to find the most practical option. Both options feature triangular holes for strength, with many, smaller holes on one and fewer, larger holes on the other. There are also multiple options for the attachment mechanism for the lid of the CanSat, which will also be tested for their durability and strength. One includes a twisting lock mechanism, whilst the other two utilise larger clips attached to the side. The lid will also feature a number of holes for parachute chords, which will be tied to attachment points towards the top of the CanSat's main body. The colour of the CanSat will be a bright, visible colour, preferably not green, so that we can locate the CanSat after launch.

Materials

Ideally, the CanSat will be 3D printed using ASA (Acrylonitrile Styrene Acrylate), due to the ease of manufacturing and availability of facilities for 3D printing. ASA has been selected due to its strength and durability, and its credibility in surviving impacts. Furthermore, it is highly weather-resistant and widely used for outdoor purposes, such as our own. If ASA is not a viable choice, ABS (Acrylonitrile Butadiene Styrene) is also a strong option, due to its similar strength and survivability.

In prototyping and testing, however, PLA (Polylactic Acid) will be used, due to its low cost and simple printing process. A group member can also facilitate printing with PLA, making it especially efficient and accessible to us.

3.2 Electrical design

For our main microcontroller/processor, we initially shortlisted two main options : The Raspberry Pi Pico and the Arduino Feather.

Raspberry Pi Pico	Arduino Feather
- Very light and low form factor - Very low cost, fitting our budget - Good support for Micropython/Circuitpython, both of which our team has experience in - Multiple different interfaces (I2C, SPI etc.) to add secondary mission sensors.	- Also light and small - More expensive - Some, but less, support for Circuitpython, C/C++ may be required - Less pins and interfaces for additional sensors beyond the primary mission. - Lots of variants like WiFi/SD Card

Although both would make good choices as they have excellent support for our chosen sensors, especially the Feather, we decided to use the Raspberry Pi Pico. Our team has far more experience with it and its supported languages, allowing us to create more ambitious and robust programs . Additionally, the pico is much easier to learn, allowing us to teach less experienced members of the group and perform potential workshops with a much wider audience.

For our temperature/pressure sensor, we have chosen the BMP 280. While potentially more advanced/powerful models are available such as the BME 280, which would allow us to measure humidity, or the BMP 390, which would allow us to take reading with much higher precision, we have still chosen the 280. High precision or humidity readings would be excessive, as the tolerances below will more than suffice for 400m, and the BMP280 also has the best compatibility with CircuitPython and the Pi Pico, again allowing us to teach others easily. Its tolerances are as follows:

Pressure range : 300 – 1100 hPa   Relative accuracy : ±0.12 hPa, equiv. to ±1 m (@25C)

Temperature range : -40 – +85 °C Temperature offset : 1.5 Pa/K, equiv. to 12.6 cm/K (@900Pa)

Datasheet used : https://cdn-shop.adafruit.com/datasheets/BST-BMP280-DS001-11.pdf

As for our telemetry, we will most likely be using the Adafruit LoRa RFM9x. The RFM9x is a long-range transceiver, enabling us to use the same module for receiving on the ground and in the CanSat, drastically decreasing the margin for error. Again, it is also completely compatible with the Pi Pico and CircuitPython. However, a major concern for using this module would be the range capabilities. Although it is technically capable of 2km in range, initial testing has found this may be unrealistic. This issue becomes more of a concern as the radio requires 5V to operate at full capacity, and 3-4V may only be possible due to space restrictions in the can. To counteract this, we will use a quarter-wavelength whip antenna on the CanSat, which resonates with the 433Mhz frequency the radio uses and allows for better transmission. However, this may still not give us the required range and signal strength, so we will carry out extensive testing to conclude as to whether utilising a different radio module is necessary.

Datasheet used:

https://cdn-learn.adafruit.com/assets/assets/000/031/659/original/RFM95_96_97_98W.pdf?1460518717

Lastly, for our secondary mission we will be using the MPU6050. The MPU is an accelerometer as well as a gyroscope, and gives separate values for each axis, making calculations much easier to perform. One critical advantage of this sensor is its very small footprint of 20x15mm, as with the BMP and radio there is very little space left. In addition, it operates at a low minimum voltage of 2.3V, works on a simple I2C interface, and once again is completely compatible with circuitpython. However, one major disadvantage is its relatively low accuracy and tolerance. A small variation in readings could change calculated values by a lot. Unfortunately, the aforementioned space restrictions prohibit us from being able to use a more powerful sensor. Additional adjustments will have to be made to compensate in the calculations, such as averaging the height with the BMP's altitude reading.

3.3 Software design

The software required for the CanSat consists of a measurement system, radio transmission system, ground system and potential post analysis systems.

We are using Thonny for the Raspberry Pi Pico due to its ease and speed of development and our group's familiarity with the interpreter and GitHub for the version control and efficient group development workflow.The LoRa radio rfm9x and the sensor bmp280 have great compatibility with CircuitPython (a python based language) which our group knows extensively and contains thorough documentation online. This makes both the language, the sensors and the microcontroller suitable for our project as opposed to an Arduino and C/C++ development of which we have little experience within our group.

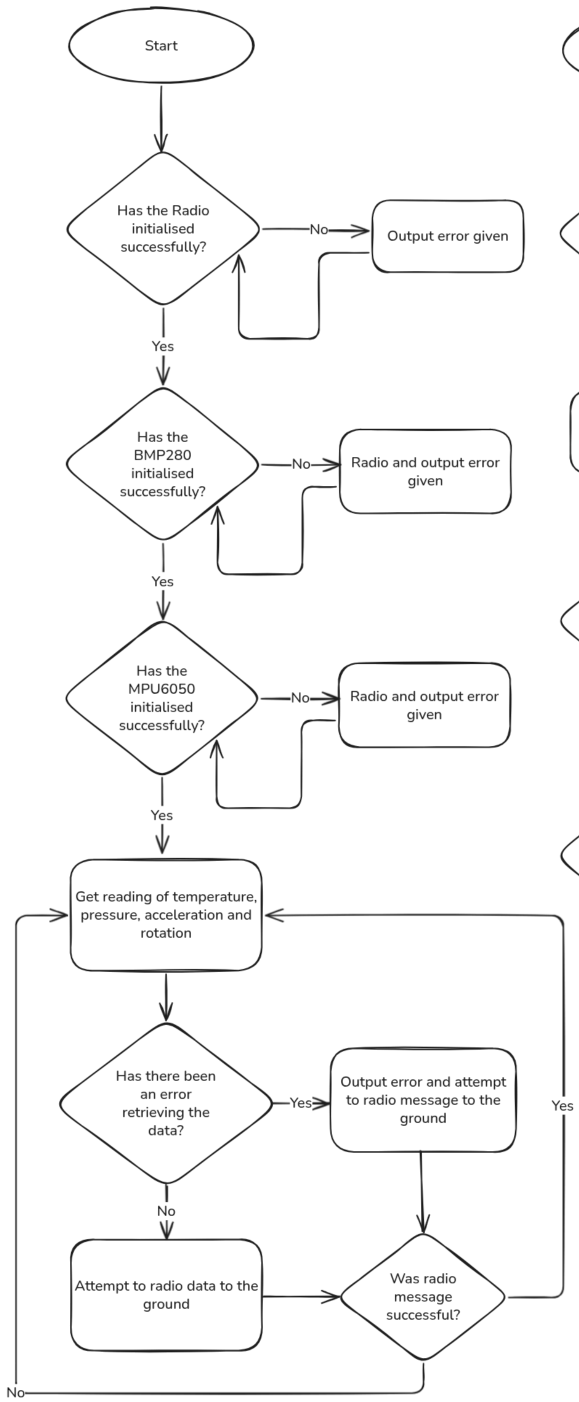
A module to develop is a radio and sensor class(es) that contains the core methods for the radio transmissions and measurements in the CanSat code which is passed as a library into the main routine where all the checks and transmissions are made.This makes the code more readable and easier to debug and track errors in the future (future-proofing code)

Other code to be developed is the receiving and storage of data in the ground station. Here we simply wait for the laptop to establish a connection to the radio and print the data Received data on temperature, air pressure, acceleration, direction and signal strength will then be formatted into a CSV file, so that we can analyse this later on.

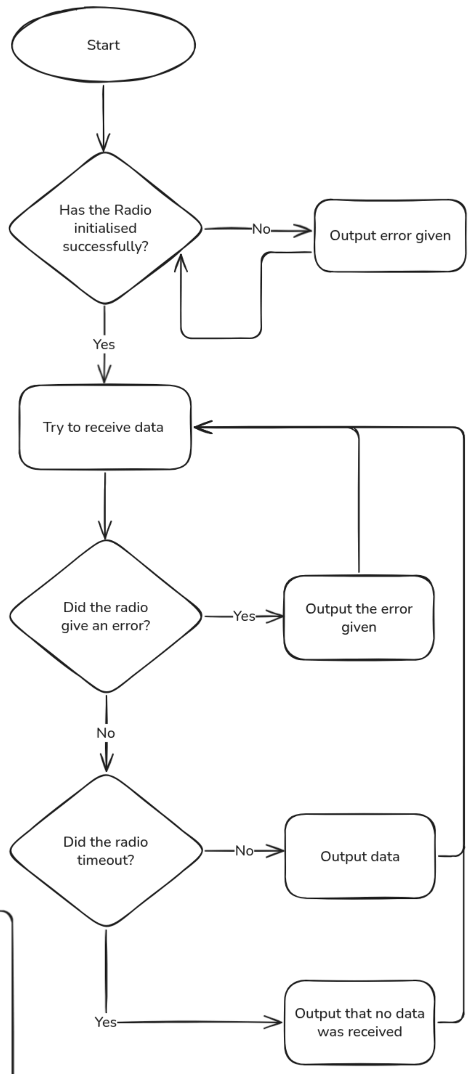
To analyse the data, we will use various python libraries. To read the data off of the CSV we will use pandas as it is easy to use, allowing us to quickly obtain and format the data we need. To perform some of the complex calculations that will be necessary for our secondary mission,, we will use numpy due to its plethora of built in, optimized functions that fit our purposes. To display our data we will use matplotlib, as it works well with numpy and has an excellent variety of 2D and 3D graphs in Python. Our team has some experience with these libraries and they give us a holistic toolkit to scientifically analyse our data.

What follows is a high-level flowchart showing the processes for the primary and secondary mission of both the CanSat and ground station:

CanSat Code:



Ground Code:



3.4 Landing and recovery system

For the CanSat*'s* landing system, we've chosen to use a hexagonal shaped parachute. This particular design offers a moderate drag coefficient (~0.75 - 0.8) and we are confident in making it accurately. The hexagonal shape can be sewn together from 6 separate triangles, each congruent to each other.

To find the cross-sectional area for the parachute, we need to consider the following constraints:

- The descent rate of the CanSat must be between 10 - 15 m/s
- The CanSat's mass must be between 300 - 350 grams
- For this calculation, as this is only an estimation, we will be taking the upper bound for the mass

For this calculation, we will be using this equation which expresses the equal weight force and drag force when the terminal velocity is reached:

$m \cdot g - \frac{1}{2} \cdot C_D \cdot \varrho \cdot A \cdot v^2 = 0$

The variables in the equation are:

m = mass of CanSat, in this case, ~350 grams

g = gravitational acceleration on Earth, ~9.8 m/s^2

C_D = drag coefficient, the total drag resistance, not directional, in this case, 0.75, taken from the lower bound in the case of slight inaccuracies in the making of the parachute

ϱ = fluid density, in this case, density of air (however only at sea level, provides a moderately valid estimation), 1.225 kg/m^3

A = cross-sectional area of the parachute

v = the required terminal velocity of the CanSat, in this case, 10 m/s

This shows the balance in drag and weight force from the 2 parts of the equation:

$m \cdot g = F_G$, the force an object exerts when accelerated by gravity

$\frac{1}{2} \cdot C_D \cdot \varrho \cdot A \cdot v^2 = F_D$, the drag force required to reach the terminal velocity of v

- The equation expresses them both being equal (one subtracting the other = 0), as the forces would be when the object moves at constant velocity

This can be rearranged to make area the subject:

$m \cdot g = \frac{1}{2} \cdot C_D \cdot \varrho \cdot A \cdot v^2$

$2 \cdot m \cdot g = C_D \cdot \varrho \cdot A \cdot v^2$

$A = 2 \cdot m \cdot g / C_D \cdot \varrho \cdot v^2$

Now we substitute the values:

$A = 2 \cdot 0.35 \cdot 9.8 / 0.75 \cdot 1.225 \cdot 10^2$

$A \approx 0.0747\ m^2$ or $747\ cm^2$

From these calculations we reach the estimated parachute design of a hexagonal shape with a side length of ~17 cm and a cross-sectional area of ~747 cm^2.

We are also not very concerned about the tangling issues when deploying the parachute, as this design does not require a massive mess of strings. Adding any extra components to aid the deployment might add extra complications and problems we have to deal with. We will be very careful in preparing the parachute to avoid as many opportunities for tangling as possible.

We also believe that the parachute would, most likely, not fail the deployment. As these early estimates show that the parachute is not small in size and it will definitely catch air during launch. However, we must also take into account the weight of the fabric of the parachute. If it is too heavy, it might not be able to spread out completely and lose stability and be less effective.

3.5 Ground support equipment

On the ground, the most important component for us is the antenna. As mentioned in the electrical design section, weak radio connection is a major risk. Therefore, for our antenna we will be using a UHF yagi antenna. This antenna is perfect as our radio modules operate at 433Mhz in the UHF band, it is very lightweight and manoeuvrable, and operates with a high gain of 7 dBi. However, the antenna is uni-directional, so we will have to track the visual trajectory of the CanSat in order to get optimal results. We will dedicate a person to this on the launch day, but this still may reduce the efficiency of the radio as it will be very difficult to track it at high altitudes.

As for other equipment on the ground, we will be using another Pi Pico connected to a laptop via USB. The pico will use another adafruit radio module to transfer data over serial to a python program that will process the data. This is outlined in the software design section. In order to ensure minimal margin for error, we will need to ensure that the radio, antenna, pico and USB cable are attached as securely as possible, as even a second of connection loss during the launch could cause large loss of data. Additionally, we will have to ensure the laptop is sufficiently charged, to maximise the reliability of our setup. The overall process of the ground equipment is outlined bellow:

1) Data is received with the antenna
2) Data is passed over a serial connection to the the laptop
3) The python program on the laptop records the data into a CSV file
4) After the launch, the CSV file is analysed and processed

3.6 Testing

What we are testing	How we will test	If the test is pa	ailed
Structural integrity	Repeatedly drop the CanSat from a height of around 1m. Inspect for loose parts and broken connections. Check if telemetry is functioning before and after the test	We have confirmed that the CanSat can likely survive the high forces that will be experienced during launch	Could lead to the destruction of the can or internal electronics. Redsign the frame to eliminate loose parts, secure wiring and perhaps increase infill on our 3D printed design
Size and weight restrictions	Measure the dimensions and mass of our CanSat to ensure that it is within the required range	No changes needed	Could be disallowed to launch Redesign the CanSat to be smaller, add ballast, redesign internal structure
Battery accessibility	Time how long it takes for someone to turn off the battery with gloves on	Requirements have been met	Could be disallowed to launch Reposition the switch, widen access holes
Battery life	Run the CanSat with the fully functioning sensor for 3 hours and observe results	Our CanSat can meet the required battery life	May lose power on the day suddenly Investigate alternative battery options with a longer life, improve code efficiency
BMP-280 temperature accuracy	Compare readings at room temperature to that of a thermometer, see if temperature increases or decreases as expected when placed outside or in front of a hairdryer.	Our temperature readings are accurate and reliable	Inaccurate readings of temperature could give us useless data Try and calibrate our sensor, replace the sensor.
BMP-280 pressure and altitude readings	Use the pressure readings to calculate altitude values and see if they are reasonable and in line with reference values. Check if altitude reading increases at the top of a tall building	We have accurate readings for pressure and altitude.	Inaccurate readings for altitude could prevent us from making any correlations. Try and calibrate our sensor, replace the sensor.
MPU 6050 accuracy	Record values when stationary and when moving. Check for drift and bias. Do the same with orientation by rotating the sensor in specific axes	We have values for acceleration and direction that we can use for our calculations	Inaccuracies in this sensor could sabotage our secondary mission. Try and remove any repeated bias or drift, calibrate the sensor, perhaps by a new sensor

What we are testing	How we will test	If the test is pa	ailed
Radio functionality	Run the radio connected to all of the sensors and observe the output from telemetry and if packets are intact	We have reliable telemetry	We may have corrupted data that cannot be analysed. Fix wiring, improve soldering on the LorA, look into the settings of the radio
Radio range	Check if the radio is fully functional at a range of at least 1km	No risk of disconnection from range being exceeded	We may lose connection due to altitude. Adjust antenna length and orientation.
Parachute length	Test if the parachute connection can withstand a force of 50N.	Confident parachute deployment	Risk of tearing and detachment. Change material of lines, attach using glue gun, improve knots
Parachute reliability	Drop the parachute from a reasonable height repeatedly and confirm that it deploys consistently	Confident parachute deployment	Risk of exceeding recommended speed. Experiment with different deployment methods.
Descent rate	Drop the parachute from a reasonable height and check if its terminal velocity is within the required limits	Requirement met	Risk of not meeting requirements. Redesign parachute based on test results to meet requirements
Descent stability	Observe the CanSat during descent and ensure that there is not excessive oscillation	Valid secondary mission data, safe descent	Unsafe descent, risk of unusable secondary mission data. Adjust attachment points, redistribute mass
Signal strength	Record the RSSI of the yagi antenna when receiving the telemetry data	Reliable telemetry	Potential loss of signal midflight. Retune antenna, identify best technique
Full mission runthrough	Run through a practice launch day step by step, with the finished CanSat, dropping it from a tall building. Ensure all functions are carried out	Practice for launch day, more confidence	Risk of failure on launch day. Streamline process, reassign roles for the day

These are other tests we shall conduct that are minor tests and need not in-depth analysis::

- Changing frequency and testing telemetry
- Check timestamps to see if readings are sent every second
- Ensure packets convert cleanly to CSV format
- Testing CanSat recovery

3.7 Overall testing for launch

We have covered these tests in the previous section

4 Outreach programme

For our outreach, we plan to take a varied approach with differing methods to promote the CanSat competition. Here are a few of our ideas:

- Social media

Our local robotics club has a fairly large social media presence that we could use to our advantage. Using some videos from both our participation in the competition last year and our current work, we could raise awareness of this competition and encourage STEM related activities

- Teaching sessions

Another idea of ours is to provide a workshop-like service at our local robotics club. We could create an event open to the public where we talk about our experiences and work related to the CanSat competition, how it has helped us, and perhaps demonstrate some sensors and the telemetry of our CanSat. This would be open to all in our local area.

- Assemblies

Assemblies in our school are conducted on various topics, and often students get opportunities to participate in them. We believe that by conducting a presentation to our school, we could spread our knowledge and skills to our peers and show the benefit of STEM competitions like CanSat. Some members of our robotics club have done similar assemblies successfully in the past, so we are confident that we can get a slot.

- Local Newspaper:

Our town has a local monthly newspaper which is read by many of its residents. Some of our team members were able to make an appearance on it last year after the regional launch, helping them advertise the competition and its benefits. We plan to use this advertising platform again this year, as we have already had success with it in the past. It also allows us to reach a lot of younger students, who may not have access to our other advertising platforms such as social media.